Long-Lived Agents, AI, and Data Quality: Working Notes

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1 Recap of the Base Model (One-Period Agents)

This section briefly summarizes the notation and key results from the base model with one-period-lived entrepreneurs.

1.1 Environment and Notation

- **Fundamentals.** An economic fundamental evolves as

$$\theta_t = \rho\theta_{t-1} + \eta_t,$$

where η_t has mean 0 and precision γ_η.

- **Payoff.** Entrepreneur i takes action a_t^i and earns

$$A_t^i = \bar{A} - (\theta_t - a_t^i)^2.$$

- **Signals.** Human labor generates a private signal with precision γ^l at idiosyncratic cost $\varphi_t^i \sim F(\cdot)$. AI generates a prediction using an inherited dataset at cost R_t.

- **AI data.** The inherited dataset is denoted Ω_{t-1}, and its precision is

$$\gamma_t = \frac{1}{\mathrm{Var}(\theta_t \mid \Omega_{t-1})}.$$

1.2 Key Results

- **Labor payoff.**

$$V^l = \bar{A} - \frac{1}{\gamma^p + \gamma^l}.$$

- **AI payoff.**

$$V^{AI} = \bar{A} - \frac{1}{\gamma^p + \gamma_t} - R_t.$$

- **Adoption cutoff.**

$$\bar{\varphi}_t(\gamma_t) = R_t + \frac{1}{\gamma^p + \gamma_t} - \frac{1}{\gamma^p + \gamma^l}.$$

Remark 1 (No Public Signal)**.** *This version of the notes sets $\gamma_S = 0$ throughout (equivalently, there is no public signal). This eliminates the discrepancy between the reduced-form private transition laws used in the theoretical sections and the transition maps used in the numerical exercise. All results below are self-contained under this assumption.*

2 Long-Lived Agents Extension

2.1 Overview

The purpose of the long-lived extension is to let current choices affect the entrepreneur's future private expertise. For the moment we focus only on steady states, not on transition dynamics or welfare.

2.2 Environment with Three Technologies

Let

$$\gamma^p = \frac{1}{\mathrm{Var}(\theta_t \mid \omega_{t-1})}$$

denote the entrepreneur's private precision at the start of the period, and let

$$\gamma = \frac{1}{\mathrm{Var}(\theta_t \mid \Omega_{t-1})}$$

denote the precision of the inherited dataset available to AI.

The entrepreneur now has three technologies:

$$N = \text{no action / no new signal}, \qquad L = \text{labor}, \qquad A = \text{AI}.$$

The no-action technology is important because it prevents the pre-AI economy from mechanically pushing everyone toward the top of the skill distribution.

Table 1: Exogenous model parameters

Parameter	Economic meaning
ρ	Persistence of the latent state.
γ_η	Precision of the innovation η_t.
β	Entrepreneur's discount factor.
γ^l	Precision of the signal produced by labor.
γ_S	Precision of the public signal observed by everyone.
w	Mean labor cost in the Gaussian numerical benchmark, where $\varphi = w + \varepsilon$.
p_A	Flow cost of using AI.
σ_φ	Standard deviation of the Gaussian labor-cost shock in the numerical benchmark, with $\varepsilon \sim N(0, \sigma_\varphi^2)$.

We do not list endogenous state variables such as γ^p or γ in this table. Throughout the note the underlying environment remains the three-action model $\{N, L, A\}$. In the Gaussian numerical benchmark we specialize the labor-cost distribution to

$$\varphi = w + \varepsilon, \qquad \varepsilon \sim N(0, \sigma_\varphi^2),$$

so the code parameter named `sigma` is exactly σ_φ, not a different reduced-form object. More generally, the three-action model can be written with an arbitrary labor-cost distribution $F(\varphi)$; the Gaussian benchmark is just the numerical specification we use here.

Current-period payoffs are

$$u_N(\gamma^p) = \bar{A} - \frac{1}{\gamma^p}, \tag{1}$$

$$u_L(\gamma^p, \varphi) = \bar{A} - \frac{1}{\gamma^p + \gamma^l} - \varphi, \tag{2}$$

$$u_A(\gamma^p, \gamma) = \bar{A} - \frac{1}{\gamma^p + \gamma} - p_A. \tag{3}$$

Private precision evolves as

$$\frac{1}{\gamma_L^{p\prime}} = \frac{\rho^2}{\gamma^p + \gamma^l} + \frac{1}{\gamma_\eta}, \qquad \frac{1}{\gamma_N^{p\prime}} = \frac{\rho^2}{\gamma^p} + \frac{1}{\gamma_\eta}. \tag{4}$$

We assume AI and no action have the same private transition,

$$\gamma_A^{p\prime} = \gamma_N^{p\prime},$$

because AI helps with the current decision through the inherited dataset but does not generate a fresh private signal for the entrepreneur.

2.3 Pre-AI Stationary Equilibrium

Before AI arrives, the entrepreneur chooses only between N and L. The Bellman equation is

$$V^0(\gamma^p) = \int_0^\infty \max\left\{u_L(\gamma^p, \varphi) + \beta V^0(\gamma_L^{p\prime}),\ u_N(\gamma^p) + \beta V^0(\gamma_N^{p\prime})\right\} dF(\varphi). \tag{5}$$

Lemma 2 (Pre-AI Labor Cutoff)**.** *There exists a cutoff rule $\bar{\varphi}_0(\gamma^p)$ such that labor is used if and only if $\varphi \leq \bar{\varphi}_0(\gamma^p)$, where*

$$\bar{\varphi}_0(\gamma^p) = \frac{1}{\gamma^p} - \frac{1}{\gamma^p + \gamma^l} + \beta\left[V^0(\gamma_L^{p\prime}) - V^0(\gamma_N^{p\prime})\right]. \tag{6}$$

Definition 3 (Pre-AI Stationary Equilibrium)**.** *A pre-AI stationary equilibrium is a value function V^0, a cutoff rule $\bar{\varphi}_0$, and an invariant distribution G^0 over γ^p such that:*

1. *V^0 solves* (5)*;*
2. *$\bar{\varphi}_0$ is optimal;*
3. *G^0 satisfies*

$$G^0(B) = \int \left[F(\bar{\varphi}_0(\gamma^p))\mathbf{1}\{\gamma_L^{p\prime}(\gamma^p) \in B\} + (1 - F(\bar{\varphi}_0(\gamma^p)))\,\mathbf{1}\{\gamma_N^{p\prime}(\gamma^p) \in B\}\right] dG^0(\gamma^p) \tag{7}$$

for every measurable set B.

Two benchmark private-precision fixed points are useful. If labor is chosen every period, private precision converges to $\gamma^{p,\ell}$ satisfying

$$\gamma^{p,\ell} = \left(\frac{\rho^2}{\gamma^{p,\ell} + \gamma^l} + \frac{1}{\gamma_\eta}\right)^{-1}. \tag{8}$$

If no fresh signal is ever acquired, private precision converges to

$$\gamma^{p,n} = \left(\frac{\rho^2}{\gamma^{p,n}} + \frac{1}{\gamma_\eta}\right)^{-1} = (1 - \rho^2)\gamma_\eta. \tag{9}$$

The second equality follows from the stationarity of the AR(1) process: even an agent who never observes any signal knows the unconditional distribution of θ_t, which has precision $(1-\rho^2)\gamma_\eta$. Note that this floor vanishes as $\rho \to 1$ (the random-walk limit).

Remark 4 (Why the Pre-AI Distribution Need Not Be Top-Heavy)**.** *The pre-AI invariant distribution is not the same object as the labor-only fixed point $\gamma^{p,\ell}$. If the cost distribution F puts enough mass on high realizations of φ, many entrepreneurs optimally skip labor repeatedly and remain closer to $\gamma^{p,n}$ than to $\gamma^{p,\ell}$. So the natural object is the full invariant distribution G^0, which can put substantial mass in the middle or bottom of the skill distribution.*

2.4 Post-AI Recursive Equilibrium

After AI arrives, the aggregate state is

$$M = (G, \gamma),$$

where G is the cross-sectional distribution of private precision and γ is inherited dataset quality. An individual entrepreneur takes M as given.

The Bellman equation is

$$V(\gamma^p; G, \gamma) = \int_0^\infty \max\left\{u_L(\gamma^p, \varphi) + \beta V(\gamma_L^{p\prime}; G', \gamma'),\ u_N(\gamma^p) + \beta V(\gamma_N^{p\prime}; G', \gamma'),\ u_A(\gamma^p, \gamma) + \beta V(\gamma_N^{p\prime}; G', \gamma')\right\} dF \tag{10}$$

where $(G', \gamma') = H(G, \gamma)$ is the perceived aggregate law of motion.

Definition 5 (Recursive Equilibrium with AI)**.** *A recursive equilibrium is a value function $V(\gamma^p; G, \gamma)$, a policy rule $a(\gamma^p, \varphi; G, \gamma) \in \{N, L, A\}$, and an aggregate law of motion $H(G, \gamma) = (G', \gamma')$ such that:*

1. *given H, the policy rule maximizes the right-hand side of* (10)*;*
2. *private precision evolves according to* (4)*;*
3. *for every measurable set B, the next cross-sectional distribution is*

$$G'(B) = \int \left[\pi^L(\gamma^p; G, \gamma)\mathbf{1}\{\gamma_L^{p\prime}(\gamma^p) \in B\} + \left(1 - \pi^L(\gamma^p; G, \gamma)\right)\mathbf{1}\{\gamma_N^{p\prime}(\gamma^p) \in B\}\right] dG(\gamma^p), \tag{11}$$

   where $\pi^L(\gamma^p; G, \gamma)$ is the probability that labor is chosen at state γ^p under the optimal policy rule;

4. *the next inherited dataset quality is*

$$\frac{1}{\gamma'} = \frac{\rho^2}{\gamma + \Phi(G, \gamma)} + \frac{1}{\gamma_\eta}, \tag{12}$$

   where

$$\Phi(G, \gamma) = \int \pi^L(\gamma^p; G, \gamma)\left(\gamma^p + \gamma^l\right) dG(\gamma^p). \tag{13}$$

Equivalently, in time notation, starting from aggregate state $M_t = (G_t, \gamma_t)$, requirements 3 and 4 say that the equilibrium law of motion is

$$G_{t+1} = H_G(G_t, \gamma_t), \qquad \gamma_{t+1} = H_\gamma(G_t, \gamma_t),$$

with H_G given by (11) and H_γ given by (12)–(13).

Definition 6 (Stationary Equilibrium with AI)**.** *A stationary equilibrium with AI is a recursive equilibrium in which*

$$H(G, \gamma) = (G, \gamma).$$

So the stationary equilibrium is a fixed point in the pair (G, γ)*.*

2.5 Steady-State Choice Structure

Because A and N have the same continuation state, the comparison between them is purely static:

$$u_A(\gamma^p, \gamma) \geq u_N(\gamma^p) \quad \Longleftrightarrow \quad p_A \leq \frac{1}{\gamma^p} - \frac{1}{\gamma^p + \gamma}. \tag{14}$$

Proposition 7 (AI Is More Attractive for Lower-Skill Agents)**.** *The right-hand side of* (14) *is decreasing in* γ^p*. Therefore, for a given inherited dataset quality* γ*, AI weakly displaces no action first among lower-expertise entrepreneurs.*

Proof. Differentiate the right-hand side of (14) with respect to γ^p:

$$-\frac{1}{(\gamma^p)^2} + \frac{1}{(\gamma^p + \gamma)^2} < 0.$$

□

Define the outside-option value

$$W_0(\gamma^p; G, \gamma) = \max\{u_N(\gamma^p), u_A(\gamma^p, \gamma)\} + \beta V(\gamma_N^{p\prime}; G', \gamma'). \tag{15}$$

Then labor is chosen if and only if

$$\varphi \leq \bar{\varphi}(\gamma^p; G, \gamma), \tag{16}$$

where

$$\bar{\varphi}(\gamma^p; G, \gamma) = \bar{A} - \frac{1}{\gamma^p + \gamma^l} + \beta V(\gamma_L^{p\prime}; G', \gamma') - W_0(\gamma^p; G, \gamma). \tag{17}$$

At a stationary equilibrium $(G', \gamma') = (G, \gamma)$, so the continuation values may be evaluated at the current aggregate state.

Remark 8 (What AI Changes in Steady State)**.** *Since AI and no action share the same private transition, AI does not create a third private-knowledge accumulation map. Instead, it changes the skill distribution by improving the outside option relative to labor. In other words, AI affects the stationary distribution by lowering the set of states for which entrepreneurs are willing to pay the labor cost.*

2.6 Stationary Distribution and Dataset Quality

Let

$$\pi^L(\gamma^p; G, \gamma) = F(\bar{\varphi}(\gamma^p; G, \gamma))$$

be the stationary probability of labor. The invariant distribution in a stationary equilibrium with AI must satisfy

$$G(B) = \int \left[\pi^L(\gamma^p; G, \gamma)\mathbf{1}\{\gamma_L^{p\prime}(\gamma^p) \in B\} + \left(1 - \pi^L(\gamma^p; G, \gamma)\right)\mathbf{1}\{\gamma_N^{p\prime}(\gamma^p) \in B\}\right] dG(\gamma^p) \tag{18}$$

for every measurable set B.

To close the stationary equilibrium, inherited dataset quality must also be constant. A natural reduced-form equation is

$$\frac{1}{\gamma} = \frac{\rho^2}{\gamma + \Phi(G, \gamma)} + \frac{1}{\gamma_\eta}, \tag{19}$$

where

$$\Phi(G, \gamma) = \int \pi^L(\gamma^p; G, \gamma)\left(\gamma^p + \gamma^l\right) dG(\gamma^p). \tag{20}$$

Only labor users contribute fresh human information to the inherited dataset in this formulation; no action and AI only reuse existing information.

Remark 9 (What We Are Solving)**.** *The stationary equilibrium problem with AI is:*

1. *solve the individual's Bellman equation at a candidate aggregate state* (G, γ)*;*
2. *compute the implied labor cutoff and invariant distribution;*
3. *compute the implied inherited dataset quality;*
4. *look for a fixed point in* (G, γ)*.*

That is the right object to understand before studying transition dynamics.

Remark 10 (Multiplicity at a Fixed γ)**.** *There are two distinct multiplicity questions here. One is whether the fixed-point map for inherited dataset quality admits multiple values of* γ *satisfying stationarity. The other is whether, even holding* γ *fixed, there may be several invariant distributions* G *satisfying* (18)*. The second possibility should not be ignored a priori. In the current reduced-form formulation, it is a property of the invariant-distribution operator rather than of the Bellman equation itself, so it has to be checked numerically or proved separately.*

2.7 Multiplicity and What to Compute First

The main steady-state question is whether the post-AI fixed-point problem admits one or several solutions. A convenient way to organize that question is to define a steady-state map:

1. guess inherited dataset quality γ;
2. solve the stationary individual problem with actions $\{N, L, A\}$;
3. compute the induced invariant distribution G_γ;
4. plug (G_γ, γ) into (19) to obtain an implied value $T(\gamma)$;
5. look for fixed points of
$$T(\gamma) = \gamma.$$

Multiple stationary equilibria correspond to multiple fixed points of this map.

Remark 11 (Numerical Objects to Plot)**.** *The first steady-state figures we want are:*

- *the pre-AI invariant distribution G^0;*
- *the post-AI invariant distribution G_γ at each stationary fixed point;*
- *the labor share*
$$x(\gamma) = \int \pi^L(\gamma^p; G_\gamma, \gamma) dG_\gamma(\gamma^p);$$
- *the AI share and no-action share among non-labor users;*
- *the fixed-point graph of $T(\gamma)$ against the 45-degree line.*

These are the objects that would tell us whether AI changes the shape of the steady-state skill distribution and whether multiplicity is actually present.

2.8 Numerical Illustration: Clear Bimodality in the Three-Action Model

The numerical section focuses only on the cleanest stationary bimodality exercise. The underlying environment remains the *three-action* model with actions $\{N, L, A\}$. For the numerical benchmark we specialize the labor-cost distribution to a Gaussian form,

$$\varphi = w + \varepsilon, \qquad \varepsilon \sim N(0, \sigma_\varphi^2),$$

and then solve the stationary equilibrium implied by that specification. The results in this subsection are produced by our MATLAB code in `bimodality_no_public_signal.m`.

The benchmark calibration is

$$\rho = 0.8, \qquad \gamma_\eta = 5, \qquad \gamma^l = 0.5, \qquad \gamma_S = 0, \qquad \beta = 0.3, \qquad w = 0, \qquad p_A = 0.169744,$$

Table 2: Benchmark calibration for the main bimodality exercise

Parameter	Value
ρ	0.8
γ_η	5
γ^l	0.5
γ_S	0
β	0.3
w	0
p_A	0.169744
σ_φ	0.02

so the implied net cost gap is $\Delta = w - p_A = -0.169744$. Here $\sigma_\varphi = 0.02$ is the dispersion of the Gaussian labor-cost shock. In the code this same object is stored in the variable named `sigma`.

Because $\gamma_S = 0$, the action-specific state transitions are exactly the private transition laws stated in (4):

$$\Psi_A(\gamma^p) = \left(\frac{\rho^2}{\gamma^p} + \frac{1}{\gamma_\eta}\right)^{-1}, \qquad \Psi_L(\gamma^p) = \left(\frac{\rho^2}{\gamma^p + \gamma^l} + \frac{1}{\gamma_\eta}\right)^{-1}.$$

This is the key advantage of setting $\gamma_S = 0$: the theoretical transition laws and the numerical transition maps are identical.

At a fixed aggregate AI quality γ, the labor-versus-AI threshold is

$$\bar{\delta}(\gamma^p; \gamma) = \frac{1}{\gamma^p + \gamma} - \frac{1}{\gamma^p + \gamma^l} + \beta \left[V(\Psi_L(\gamma^p); \gamma) - V(\Psi_A(\gamma^p); \gamma)\right].$$

Given the Gaussian perturbation, labor is chosen with probability

$$\pi^L(\gamma^p; \gamma, \Delta, \sigma_\varphi) = \Phi_N\left(\frac{\bar{\delta}(\gamma^p; \gamma) - \Delta}{\sigma_\varphi}\right),$$

where Φ_N denotes the standard normal CDF.

This is still a three-action equilibrium construction. In the pre-AI economy, AI is unavailable, so agents choose between L and N. In the post-AI benchmark, the calibrated equilibrium satisfies the bound under which AI weakly dominates inaction, so no agent chooses N in equilibrium and the active choice set collapses to $\{L, A\}$. The code exploits this equilibrium restriction, but the economic environment is still the original three-action model.

The aggregate AI-quality fixed point is computed from labor-generated human data using

$$\Phi(G, \gamma) = \int \pi^L(\gamma^p; \gamma)(\gamma^p + \gamma^l)\, dG(\gamma^p).$$

The numerical algorithm is:

1. guess an aggregate inherited data quality γ;
2. solve the Bellman equation at that γ by value-function iteration;
3. compute the implied labor probability $\pi^L(\gamma^p;\gamma)$ on the private-precision grid;
4. iterate the cross-sectional distribution forward under the two private transition maps until the invariant distribution is reached;
5. compute the implied human-information flow $\Phi(G,\gamma)$;
6. update γ from the stationary aggregate law of motion and search for a fixed point by bisection.

The private-precision grid is chosen to bracket the two deterministic attraction points implied by these transition maps. The *follower* fixed point solves

$$\gamma^{p,f} = \Psi_A(\gamma^{p,f}),$$

which under the present calibration gives

$$\gamma^{p,f} = (1-\rho^2)\gamma_\eta = 1.8000.$$

This is exactly the unconditional precision of the AR(1) process—even an agent who never observes any signal knows the unconditional distribution of θ_t, which provides this floor. As $\rho \to 1$ the floor vanishes.

The *expert* fixed point solves

$$\gamma^{p,e} = \Psi_L(\gamma^{p,e}),$$

which gives

$$\gamma^{p,e} \approx 2.3595.$$

The numerical grid covers

$$\gamma^p \in [0.8,\ 4.2]$$

with 1201 equally spaced nodes. Bellman iteration is run to tolerance 10^{-12}, invariant-distribution iteration to tolerance 10^{-13}, and the fixed point in γ is solved to within about 10^{-7}.

Under this calibration, our code finds:

$$\sigma_\varphi = 0.0005: \quad \gamma^* \approx 2.9580, \quad x^* \approx 0.5866,$$

$$\sigma_\varphi = 0.02: \quad \gamma^* \approx 2.7235, \quad x^* \approx 0.3975,$$

$$\sigma_\varphi = 0.05: \quad \gamma^* \approx 2.7905, \quad x^* \approx 0.4712.$$

The economically most useful case is the middle one,

$$\sigma_\varphi = 0.02.$$

For this case, the stationary equilibrium computed by our code is

$$\gamma^* = 2.723471283301, \qquad x^* = 0.397504562806.$$

At $\sigma_\varphi = 0.0005$ the distribution is still close to near-deterministic polarization. At $\sigma_\varphi = 0.02$ the two masses broaden into a clear bimodal stationary density. At $\sigma_\varphi = 0.05$ the distribution remains bimodal but becomes more diffuse.

We also checked whether a fixed value of γ can support multiple invariant distributions in the clear bimodal case. Holding

$$\sigma_\varphi = 0.02, \qquad \gamma = 2.723471283301$$

fixed, we iterated the invariant-distribution operator from three very different initial conditions: uniform, follower-point-mass, and expert-point-mass. All three converged to the same invariant distribution up to numerical precision. The largest sup-norm gap across the three resulting distributions was about 3.6×10^{-12}. So in this positive-variance case we do *not* find evidence of multiple ergodic distributions associated with the same γ.

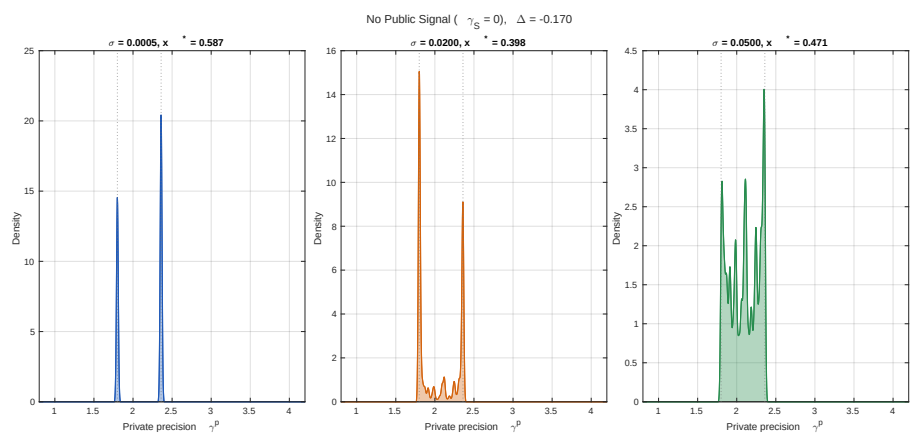

Figure 1: Three-action benchmark with $\gamma_S = 0$, evaluated in a region where post-AI equilibrium uses only L and A. Left: near-deterministic polarization at $\sigma_\varphi = 0.0005$. Middle: a clear smooth bimodal stationary distribution at $\sigma_\varphi = 0.02$. Right: a broader bimodal distribution at $\sigma_\varphi = 0.05$. The vertical lines mark the follower fixed point $\gamma^{p,f} = (1-\rho^2)\gamma_\eta = 1.8$ and the expert fixed point $\gamma^{p,e} \approx 2.36$.

As a grid-robustness diagnostic, we reran the $\sigma_\varphi = 0.02$ case on grids with 801, 1601, and 3201 nodes. The equilibrium objects barely move:

$$|\gamma^*_{801} - \gamma^*_{3201}| \approx 1.2 \times 10^{-5}, \qquad |x^*_{801} - x^*_{3201}| \approx 8.8 \times 10^{-6}.$$

So the stationary equilibrium itself is stable to grid refinement. However, the raw density oscillations do *not* disappear as the grid is refined. This is expected: conditional on choosing labor or AI, next period's γ^p is mapped almost

deterministically by the two laws of motion, and the cross-sectional measure is repeatedly folded through those two maps. That piecewise-deterministic structure creates small ripples in the invariant density even when the broad envelope is clearly bimodal.

2.9 Pre-AI versus Post-AI Comparison

The pre-AI economy is one in which AI technology is not available, or equivalently the cost of AI p_A is prohibitively high. Agents choose only between labor (L) and no action (N). Importantly, the aggregate dataset quality γ is *not* zero in the pre-AI economy: human labor generates data, and γ is pinned down by the stationary version of (19) with all agents contributing human data. Since γ enters only the AI payoff, it does not affect any individual's decision in the pre-AI world, but it is a well-defined positive object determined in the background by human information flows. Under the benchmark calibration, $\gamma_0 \approx 3.3$.

Because every agent faces only the L-versus-N choice and the static gain from labor $\frac{1}{\gamma^p} - \frac{1}{\gamma^p+\gamma^l}$ is always positive, the dynamic incentive reinforces investment: almost all agents invest in labor ($x^*_{\text{pre}} \approx 1.000$), and the stationary distribution G_0 is concentrated near the expert fixed point $\gamma^{p,e} \approx 2.36$.

When AI becomes available at a moderate price ($p_A \approx 0.170$), the post-AI stationary equilibrium features $\gamma^* \approx 2.72$ and a labor share of only $x^*_{\text{post}} \approx 0.398$. The formerly unimodal, expert-concentrated pre-AI distribution splits into a bimodal post-AI distribution: a mass of "followers" accumulates near $\gamma^{p,f} = 1.80$ (agents who switched from labor to AI and lost private expertise), while a smaller mass of persistent "experts" remains near $\gamma^{p,e} \approx 2.36$.

Figure 2 displays the two distributions side by side.

2.10 Verification That Inaction Is Not Used Post-AI

In the three-action model, no agent chooses inaction post-AI if AI weakly dominates N at every relevant expertise level. From (14), AI dominates N at expertise γ^p iff $p_A \leq \frac{\gamma}{\gamma^p(\gamma^p+\gamma)}$. Since this is decreasing in γ^p, the binding constraint is at the expert:

$$p_A \leq \frac{\gamma^*}{\gamma^{p,e}(\gamma^{p,e} + \gamma^*)}. \tag{21}$$

In the benchmark calibration ($\gamma^* \approx 2.72$, $\gamma^{p,e} \approx 2.36$), the bound evaluates to 0.227, comfortably above $p_A = 0.170$.

For labor versus AI, only the net cost gap $w - p_A$ matters. If w is too high relative to p_A, no one invests and the economy deskills. If w is too low, everyone invests and AI is irrelevant. An interior equilibrium with polarization requires $w - p_A$ in an intermediate range.

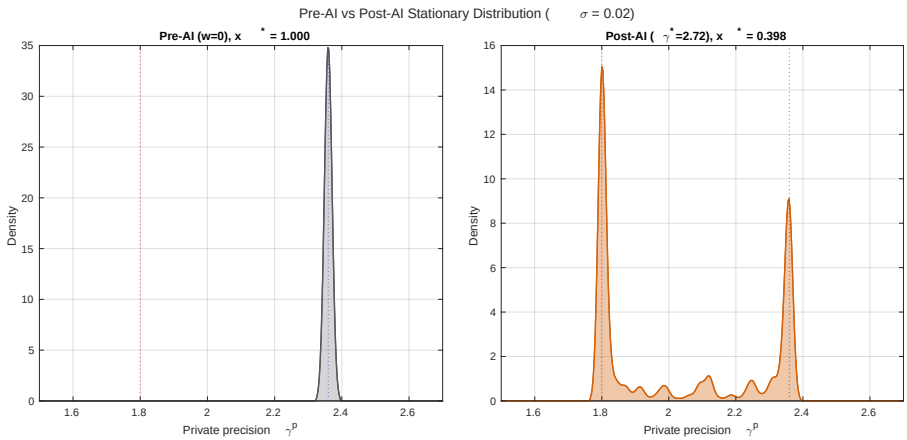

Figure 2: Pre-AI (left) versus post-AI (right) stationary distributions with $\gamma_S = 0$ and $\sigma_\varphi = 0.02$. Pre-AI: nearly all agents invest in labor ($x^* \approx 1.000$), producing a unimodal distribution near the expert fixed point. Post-AI ($p_A \approx 0.170$): the labor share falls to 40%, and the distribution becomes bimodal with modes near $\gamma^{p,f}$ and $\gamma^{p,e}$. Dashed vertical lines mark the two fixed points.

3 Competitive Equilibrium: Transition Dynamics and Welfare

3.1 Transition Dynamics

We compute the perfect-foresight transition from the pre-AI stationary equilibrium to the post-AI stationary equilibrium using backward-forward iteration on the aggregate data quality path $\{\gamma_t\}$. At $t = 0$, AI becomes available at price $p_A \approx 0.170$. The initial state is the pre-AI distribution G_0 (concentrated near $\gamma^{p,e}$) and data quality $\gamma_0 \approx 3.29$. The terminal state is the post-AI stationary equilibrium with $\gamma^* \approx 2.72$.

The algorithm iterates between a backward pass (solving time-varying Bellman equations from $t = T$ to $t = 0$ given a candidate γ path) and a forward pass (evolving the distribution and recomputing γ_{t+1} from the law of motion). The γ path is updated with damping $\alpha = 0.15$ until $\max_t |\gamma_t^{\text{new}} - \gamma_t^{\text{old}}| < 10^{-5}$.

With $T = 30$ periods, the algorithm converges in 67 iterations.

Results. Data quality declines monotonically from $\gamma_0 = 3.29$ to $\gamma^* = 2.72$. The labor share drops sharply from $x_0 \approx 1.00$ to 0.59 at $t = 0$ (when AI arrives), then gradually declines to $x^* \approx 0.40$. Most of the adjustment occurs in the first 10 periods; the economy is essentially at the new steady state by $t = 20$.

t	γ_t	x_t
0	3.288	0.585
1	3.040	0.557
2	2.950	0.518
5	2.825	0.453
10	2.756	0.415
20	2.726	0.399
30	2.724	0.398

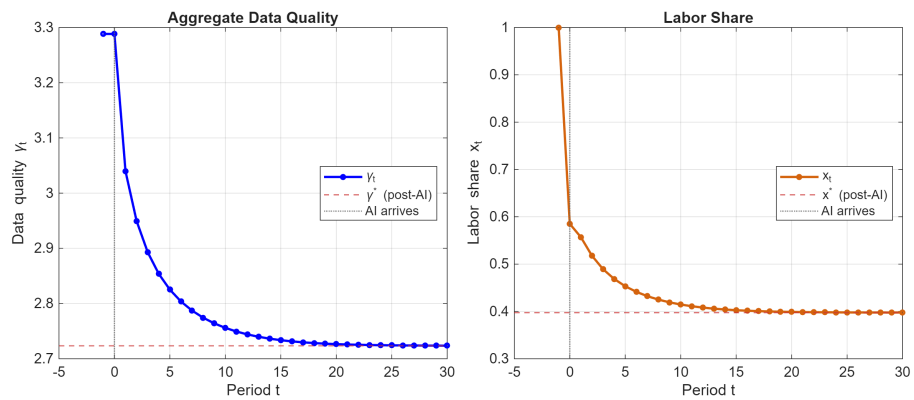

Figure 3: Transition dynamics after AI introduction ($p_A \approx 0.170$). Left: aggregate data quality γ_t declines from 3.29 to 2.72 as human data generation falls. Right: the labor share drops from near 1.00 to 0.40. Dashed lines mark the pre-AI and post-AI steady-state values.

3.2 Welfare

Welfare measure. Welfare means the lifetime payoff of a representative entrepreneur, averaged across the cross-sectional distribution of entrepreneurs. In every case we use the *actual* per-period payoffs from the model:

$$\text{if labor is used:} \qquad u_L = -\frac{1}{\gamma^p + \gamma^l} - \varphi,$$

$$\text{if AI is used:} \qquad u_A = -\frac{1}{\gamma^p + \gamma} - p_A,$$

$$\text{if no action is used:} \qquad u_N = -\frac{1}{\gamma^p}.$$

So welfare is simply: current payoff plus discounted future payoffs, averaged over agents. In the post-AI benchmark, inaction is not used in equilibrium, so welfare is computed from the entrepreneur's choice between labor and AI. In the pre-AI economy, AI is unavailable, so welfare is computed from the choice between labor and no action.

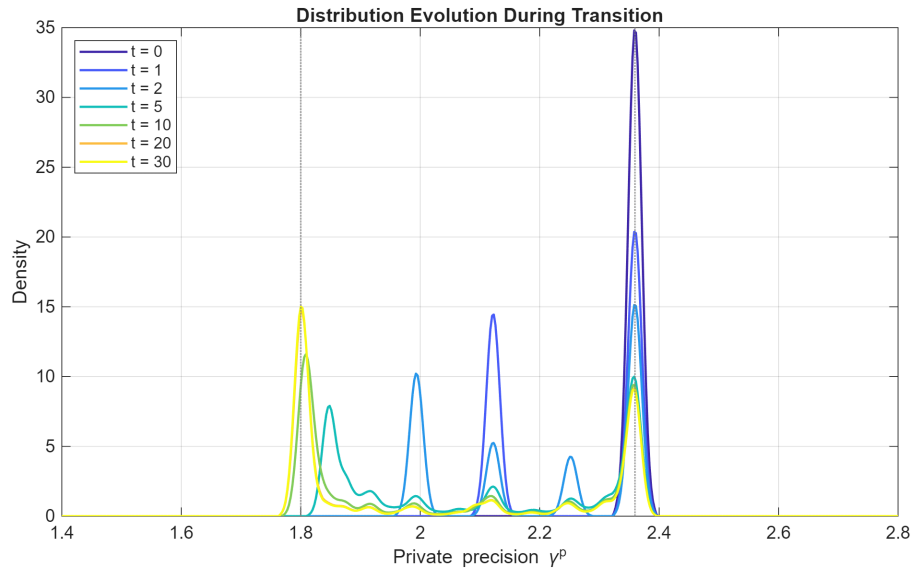

Figure 4: Evolution of the skill distribution during the transition. The pre-AI unimodal distribution (concentrated near $\gamma^{p,e}$) gradually splits into a bimodal post-AI distribution with modes at $\gamma^{p,f}$ and $\gamma^{p,e}$.

To avoid notation overload, the object we report is

$$W = \int V(\gamma^p)\, dG(\gamma^p),$$

where $V(\gamma^p)$ is the entrepreneur's lifetime value starting from current private expertise γ^p, and G is the cross-sectional distribution of private expertise in the population. So welfare is the value function averaged over the population. The only difference across the three cases below is which distribution and which value function are being averaged.

Results. We compare average welfare across three scenarios:

- **Pre-AI**: $W_{\text{pre}} = \int V^0(\gamma^p)\, dG_0(\gamma^p) = -0.4996$.
- **At AI arrival**: $W_{\text{arrival}} = \int V_0(\gamma^p; \text{transition})\, dG_0(\gamma^p) = -0.4924$, where V_0 is the value function at $t = 0$ under the perfect-foresight transition path.
- **Post-AI steady state**: $W_{\text{post}} = \int V^*(\gamma^p)\, dG^*(\gamma^p) = -0.5341$.

AI introduction raises average welfare at the moment of arrival by approximately 1.4%. This modest gain reflects the fact that, at the calibrated price $p_A = 0.170$, the cost of AI nearly offsets its precision benefit: for an expert

($\gamma^p \approx 2.36$), the MSE gain from AI is $1/(\gamma^p+\gamma^l) - 1/(\gamma^p+\gamma) = 0.350 - 0.177 = 0.173$, barely above p_A.

More strikingly, the *steady-state* CE welfare (-0.534) is *lower* than pre-AI welfare (-0.500). As agents substitute away from labor, human data generation falls, data quality deteriorates, and the long-run productivity of AI declines. The CE steady state is worse than the pre-AI economy because the data externality is not internalized: each agent ignores the effect of their choice on aggregate data quality γ. The welfare gap between pre-AI and post-AI steady state ($\approx 6.9\%$) quantifies the long-run cost of the data externality under laissez-faire.

4 Social Planner's Problem

4.1 General Formulation

The planner chooses a sequence of labor cutoff functions $\{\bar{\varphi}_t^{SP}(\gamma^p)\}_{t=0}^{\infty}$ to maximize discounted aggregate welfare, internalizing the effect of current allocations on future data quality. Agent (γ^p, φ) uses labor if $\varphi \leq \bar{\varphi}_t^{SP}(\gamma^p)$ and uses the better of AI or no action otherwise.

The planner's state is (G, γ). The recursive problem is

$$V^{SP}(G,\gamma) = \max_{\bar{\varphi}(\cdot)} \left\{ \Pi(G,\gamma;\bar{\varphi}) + \beta\, V^{SP}(G',\gamma') \right\}, \tag{22}$$

where the aggregate period payoff is

$$\Pi(G,\gamma;\bar{\varphi}) = \int \left[\int_0^{\bar{\varphi}(\gamma^p)} \big(u_L(\gamma^p) - \varphi\big)\, dF(\varphi) + \big(1 - F(\bar{\varphi}(\gamma^p))\big)\, \max\big\{u_A(\gamma^p,\gamma),\ u_N(\gamma^p)\big\} \right] dG(\gamma^p), \tag{23}$$

and the aggregate laws of motion are exactly those of the competitive economy:

$$G' = \mathcal{T}_G(G,\bar{\varphi}), \tag{24}$$

$$\gamma' = \left(\frac{\rho^2}{\gamma + \Phi(G,\bar{\varphi})} + \frac{1}{\gamma_\eta} \right)^{-1}, \qquad \Phi(G,\bar{\varphi}) = \int F\big(\bar{\varphi}(\gamma^p)\big)\big(\gamma^p + \gamma^l\big)\, dG(\gamma^p). \tag{25}$$

The distribution transition $\mathcal{T}_G$ maps each γ^p to $\gamma_L^{p\prime}$ with probability $F(\bar{\varphi}(\gamma^p))$ and to $\gamma_N^{p\prime}$ otherwise, the same law as in the competitive economy.

4.2 First-Order Condition and the Data Externality

Taking the first-order condition of (22) with respect to $\bar{\varphi}(\gamma^p)$ at a given γ^p:

$$\underbrace{\big[u_L(\gamma^p) - \bar{\varphi}^{SP} - r(\gamma^p,\gamma)\big]}_{\text{marginal agent's private net gain}} + \beta \underbrace{\frac{\partial V^{SP}}{\partial \gamma'} \frac{\partial \gamma'}{\partial \Phi} (\gamma^p + \gamma^l)}_{\text{social value of data generated}} = 0, \tag{26}$$

where $r(\gamma^p, \gamma) = \max\{u_A(\gamma^p, \gamma),\ u_N(\gamma^p)\}$ is the outside option. In the CE, only the first term appears and equals zero at $\bar{\varphi}^{CE}$. The planner adds a second term: the shadow value of data generated by the marginal labor user. Since this term is positive, the planner sets $\bar{\varphi}^{SP}(\gamma^p) > \bar{\varphi}^{CE}(\gamma^p)$, i.e., *more labor investment* than the CE. The wedge can be implemented as a tax on AI or subsidy to labor.

4.3 Stationary Planner's Problem

The planner directly chooses the agents' decision rule—a cutoff function $\bar{\varphi}^{SP}(\gamma^p)$—to maximize ex-ante steady-state welfare. Given a candidate cutoff, the stationary objects (G^{SP}, γ^{SP}) are determined by the same fixed-point conditions as the CE ((24)–(25)). The individual's value function under the planner's policy satisfies

$$V^{SP}(\gamma^p) = \mathbb{E}_\varphi\big[\mathbf{1}\{\varphi \le \bar{\varphi}^{SP}\}\big(u_L - \varphi + \beta V^{SP}(\gamma_L^{p\prime})\big) + \mathbf{1}\{\varphi > \bar{\varphi}^{SP}\}\big(u_A + \beta V^{SP}(\gamma_A^{p\prime})\big)\big]\,.$$

The planner's objective is

$$\max_{\bar{\varphi}^{SP}(\cdot)} W^{SP} = \int V^{SP}(\gamma^p)\, dG^{SP}(\gamma^p), \tag{27}$$

where G^{SP} is the invariant distribution induced by $\bar{\varphi}^{SP}$.

After solving for the planner's allocation, we compare:

1. Decision rules: $\bar{\varphi}^{SP}(\gamma^p)$ versus $\bar{\varphi}^{CE}(\gamma^p)$.
2. Steady-state objects: data quality γ^{SP} versus γ^{CE}, labor share x^{SP} versus x^{CE}, and the skill distributions G^{SP} versus G^{CE}.
3. Welfare: W^{SP} versus W^{CE}.

4.4 Numerical Results

We parameterize the planner's problem by a single shadow price $\mu \ge 0$ on aggregate data quality, yielding the augmented threshold $\bar{\varphi}^{SP}(\gamma^p) = \bar{\varphi}^{CE}(\gamma^p) + \beta\mu(\gamma^p + \gamma^l)$. We search over μ to maximize steady-state welfare. The welfare W^{SP} is evaluated using the *physical* value function (including p_A), not the augmented one used for policy computation.

	γ	x	W
Pre-AI	3.288	1.000	−0.4996
CE post-AI	2.723	0.398	−0.5341
Planner	3.286	0.997	−0.4996

The optimal shadow price is $\mu^* = 0.054$, yielding a welfare gain of 6.5% over the CE. The planner's steady state has $x \approx 1$ (nearly all labor) and $\gamma \approx \gamma_{\text{pre}}$

(near pre-AI data quality), so welfare approximately equals the pre-AI level. In other words, the planner nearly fully reverses the damage caused by the data externality: the unregulated CE erodes data quality and makes the economy *worse off* than before AI, while the planner restores the pre-AI outcome by internalizing the social value of human data.

Figure 5 helps interpret this result. The horizontal axis is the stationary labor share x induced by a given planner shadow price μ, and the vertical axis is the corresponding post-AI steady-state welfare W. The blue point is the competitive equilibrium, the red point is the best post-AI planner allocation within this policy family, and the dashed green line marks pre-AI welfare. In this calibration, steady-state welfare rises almost monotonically with the labor share and then flattens very near the pre-AI level. So the planner's near-ban on AI is not coming from a knife-edge spike in the objective; it reflects a broad region in which more labor preserves data quality and steadily improves long-run welfare.

Figure 5: Post-AI steady-state welfare as a function of the stationary labor share induced by the planner wedge. The blue point is the competitive equilibrium, the red point is the best post-AI planner allocation in this policy family, and the dashed green line marks pre-AI welfare.

We also checked the planner optimum with an alternative numerical method. Instead of the scan-plus-golden-section routine in the main code, we solved

$$\max_{\mu \in [0,0.1]} W(\mu)$$

using MATLAB's `fminbnd`. This independent search gives $\mu^*_{\text{alt}} = 0.05386$, $\gamma^*_{\text{alt}} = 3.2860$, $x^*_{\text{alt}} = 0.9968$, and $W^*_{\text{alt}} = -0.499551$, essentially the same optimum as the baseline planner computation.

It is important, however, not to overstate what this verifies. The scalar μ approach is exact only if the planner's continuation value can be summarized by a single shadow value on next period's aggregate data quality γ'. In the full planner problem, changing today's cutoff rule also changes the entire next-period expertise distribution G', not only the scalar γ'. So the μ method is a structured restriction on the planner's policy class, not a proof that the unrestricted planner problem is one-dimensional.

To check whether that restriction is driving the result, we solved a more flexible approximation to the stationary planner problem. Instead of restricting the planner to the μ family, we let the planner directly choose the cutoff rule $\bar{\varphi}(\gamma^p)$ at five nodes in γ^p space and interpolate linearly between them. This is still a finite- dimensional approximation, but it is much closer to the unrestricted problem because the planner is choosing the decision rule itself rather than a single shadow-price parameter.

The result is very close to the μ solution. The flexible node-based planner gives

$$\gamma^{SP}_{\text{flex}} = 3.2853, \qquad x^{SP}_{\text{flex}} = 0.9958, \qquad W^{SP}_{\text{flex}} = -0.499547.$$

The corresponding μ-rule solution on the same grid gives

$$\gamma^{SP}_{\mu} = 3.2861, \qquad x^{SP}_{\mu} = 0.9969, \qquad W^{SP}_{\mu} = -0.499551.$$

So the welfare difference is negligible: the two methods differ by only about 4×10^{-6} in welfare. Numerically, this is strong evidence that the near-ban on AI is not an artifact of the scalar-μ restriction in this benchmark.

The planner's welfare exceeds the pre-AI level by a small margin (0.03%) because it optimally allows a tiny fraction of agents (0.3%) to use AI when they draw a sufficiently favorable cost shock. A complete ban on AI ($x = 1$ exactly) would recover exactly $W_{\text{pre}} = -0.4996$.

The implementation question—what tax or subsidy on AI replicates the planner's allocation in a decentralized setting—is addressed after the comparison.

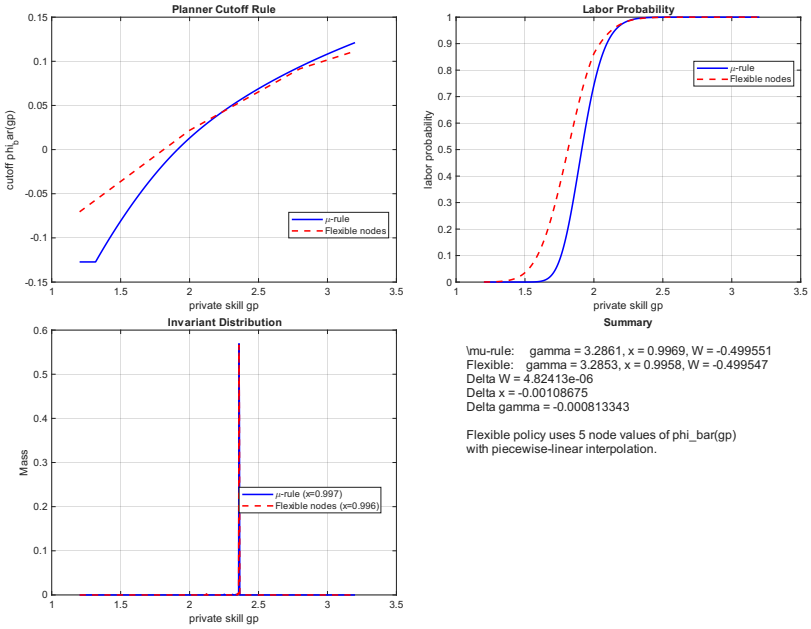

Figure 6: Comparison of the scalar-μ planner rule and a more flexible planner that directly chooses the cutoff function at five γ^p nodes. The flexible direct-cutoff solution yields almost the same labor probabilities, stationary distribution, and welfare as the μ rule.